INCH-POUND

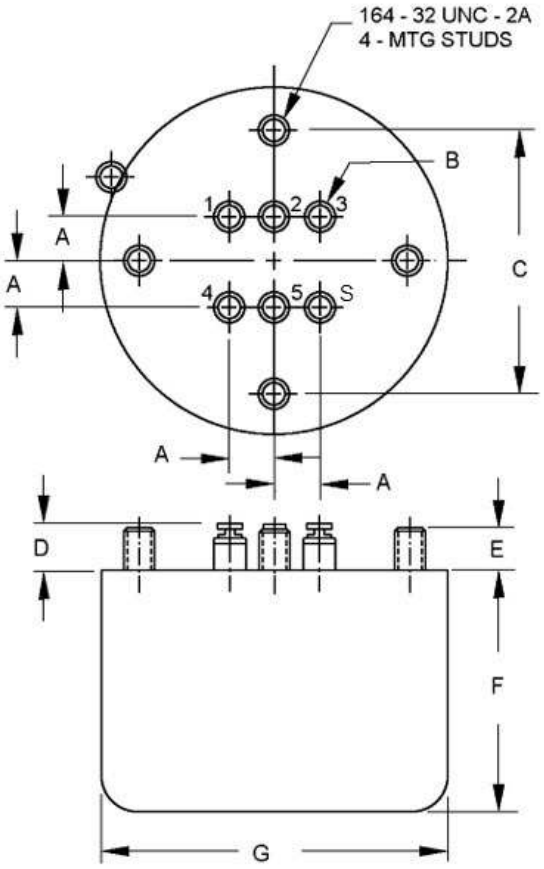
MIL-PRF-27/66E
AMENDMENT 2
2 January 2025
SUPERSEDING
MIL-PRF-27/66E
AMENDMENT 1
26 May 2016

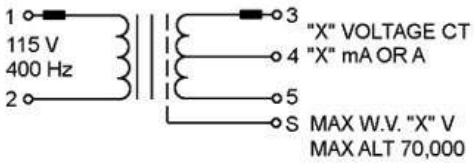
PERFORMANCE SPECIFICATION SHEET

TRANSFORMERS, POWER, 250 VOLT AMPERES, 400 HZ, ELECTROSTATIC SHIELDED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the transformers described herein shall consist of this specification and MIL-PRF-27.





(SEE TABLE I FOR VALUE OF "X"

CIRCUIT DIAGRAM AND MARKING

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Tolerance on terminal position dimensions is ±.015 (0.38 mm).
4. Marking shall be on the side of the case.
5. Secondary electrical values shall be marked as specified in table I, as applicable.

AMSC N/A
FSC 5950

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FIGURE 1. Dimensions and configuration.

Ltr	Dimensions	
	Inches	mm
A	.375	9.52
B	.250	6.35
C	2.250 MTG SQ	57.15 MTG SQ
D	.390 MAX	9.91 MAX
E	.344 ±.031	8.74 ±0.79
F	2.000 ±.031	50.80 ±0.79
G	2.875 ±.031	73.02 ±.031

FIGURE 1. Dimensions and configuration – continued.

REQUIREMENTS:
(When numbers in parentheses i.e. (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings: See table I.

Primary voltage (1-2): 115 volts ±10 percent, 400 Hz ±5 percent.

Design and construction:

Dimensions and configuration: See figure 1.

Weight: 30 ounces, maximum.

Duty cycle: Continuous.

Case: Metal enclosed.

Material: Steel.

Altitude: 70,000 feet, maximum.

Operating temperature range: -55°C to +130°C maximum.

Terminals: Feed thru.

Terminal height: .390 inch (9.91 mm), maximum.

Terminal strength: MIL-STD-202-211, test condition A, 5 pounds.

Dielectric withstanding voltage:

At sea level: 2.8 times the maximum working voltage specified in table I.

At reduced barometric pressure: 1.25 times the maximum working voltage specified in table I.

Electrical characteristics:

No load: With 115 volts, 400 Hz across (1-2): Current in (1-2) shall not exceed 125 mA rms.

Rated load: With 115 volts, 400 Hz across (1-2) and a resistive load with current as specified in table I in (3-5), the voltage across (3-5) shall be as specified in table I.

Efficiency: 95 percent minimum at rated load.

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Electrostatic shielding: The voltage ratio shall be a minimum of 5 to 1 at 20 kHz.

Polarity: Additive with terminals 2 and 3 connected.

Regulation: $\frac{Voltage\ (no\ load) - Voltage\ (rated\ load)}{Voltage\ (rated\ load)} \quad x \quad 100$

Shall not exceed 2 percent.

Phase shift: 1.0 electrical degree between primary and secondary voltage at rated load, maximum.

Temperature rise: 55°C, maximum with 115 volts, 400 Hz across (1-2) at ambient temperature of 75°C.

Vibration: MIL-STD-202-204, test condition D.

Marking location: See figure 1.

VERIFICATION:

Extent of qualification:

Qualification testing and approval to M27/49-44 A or B, M27/51-44 A or B, M27/53-44 A or B, M27/56-44 and M27/66-27 shall be sufficient to grant qualification approval to MIL-PRF-27/48 through MIL-PRF-27/66 inclusive, all parts.

Qualification testing and approval to M27/66-27 shall be sufficient to grant qualification approval to M27/66-01 through M27/66 -27.

Part or Identifying Number (PIN) example: M27/66-(dash number from table I).

TABLE I. Electrical ratings.

Dash number	Secondary voltage (3-5) (V rms) ±3%	Secondary current (3-5) (amps rms)	Working voltage (V rms max)
01	26	9.62	325
02	28	8.93	325
03	30	8.33	325
04	33	7.58	325
05	36	6.94	325
06	39	6.41	325
07	43	5.81	325
08	47	5.32	325
09	51	4.90	325
10	56	4.46	325
11	62	4.03	325
12	68	3.68	325
13	75	3.33	325
14	82	3.05	325
15	91	2.75	325
16	100	2.50	325
17	110	2.27	325
18	115	2.17	325
19	120	2.08	325
20	130	1.92	325
21	150	1.66	325
22	180	1.39	325
23	200	1.25	440
24	220	1.14	440
25	240	1.04	440
26	270	0.93	580
27	300	0.83	580

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Referenced documents. In addition to MIL-PRF-27, this document references the following:

MIL-STD-202-204 MIL-STD-202-211

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA - CC

Review activities:
Army – CR4, MI
Navy – AS, MC, OS
Air Force – 19

Preparing activity:
DLA - CC

(Project 5950-2025-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of this information above using the ASSIST Online database at https://assist.dla.mil/.

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